

Variational Autoencoder on Taiwan Cancer Registry
84,161 patients · 46 sites · 2003-2020

Cohort: 78,442 patients · 37 cancer sites (≥ 30 patients each)
Multi-cancer patients: 4,052

Model: VAE | Latent dim = 12 | $\beta = 1.0$ | Training stopped at epoch 78
Best validation ELBO: 2.7577

Active latent dimensions: see Fig 2b (KL > 0.1 threshold)
Novel axes identified (not matching UADT or hormonal): 0

Cohort Description

Fig 1a: Cancer site frequency (all registry sites ≥ 30 patients)

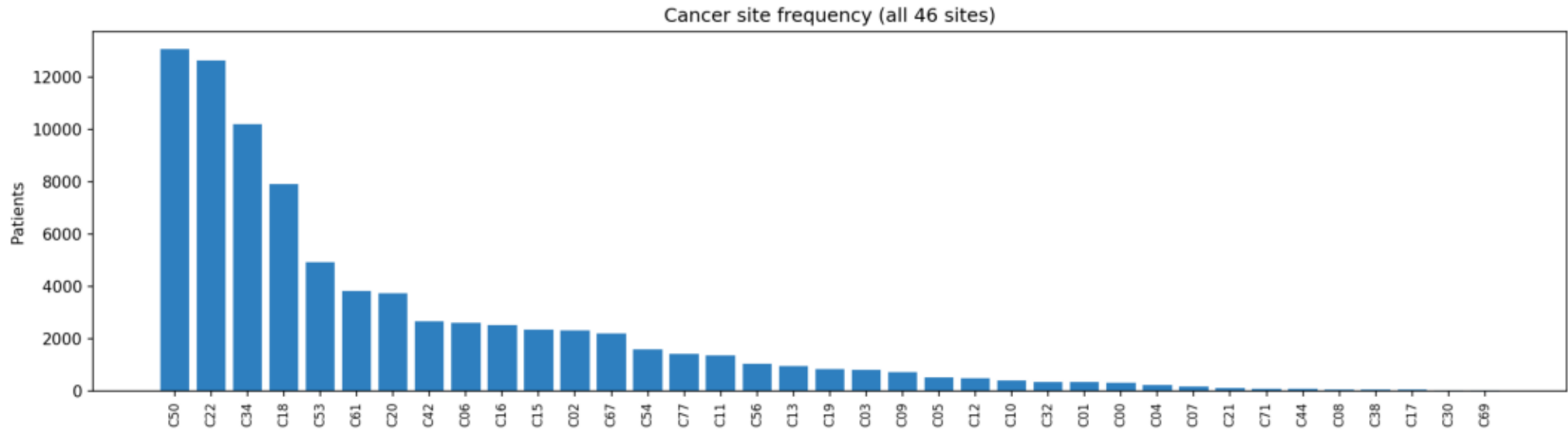
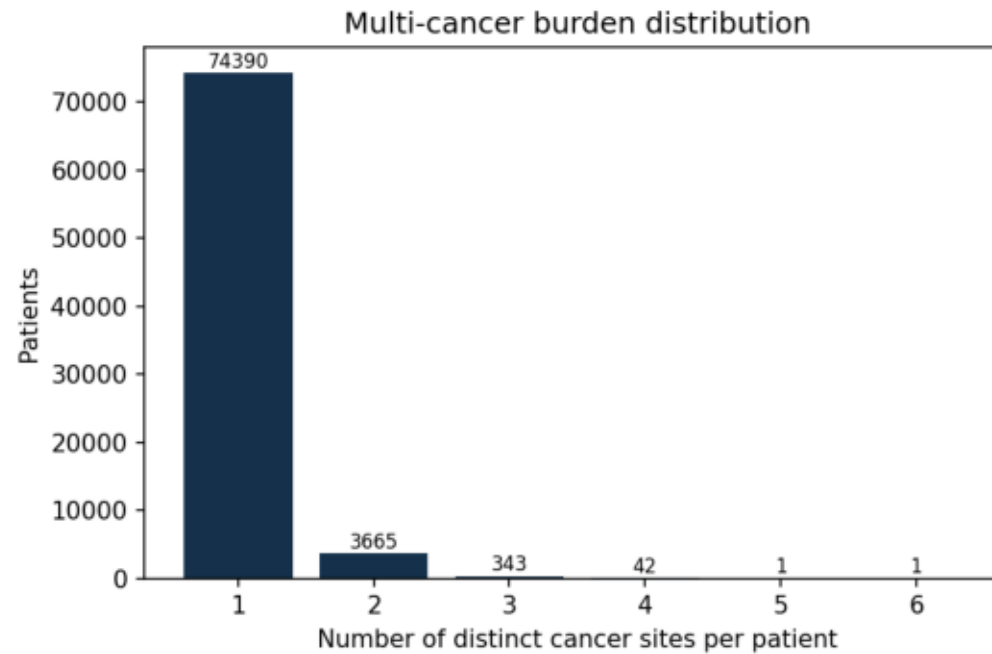


Fig 1b: Multi-cancer burden per patient



VAE Training Diagnostics

Fig 2a: Training curve

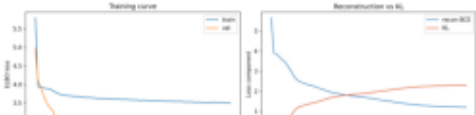


Fig 2b: KL per latent dim (active dims in blue)

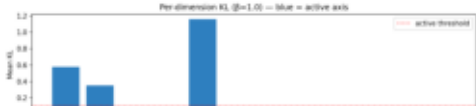


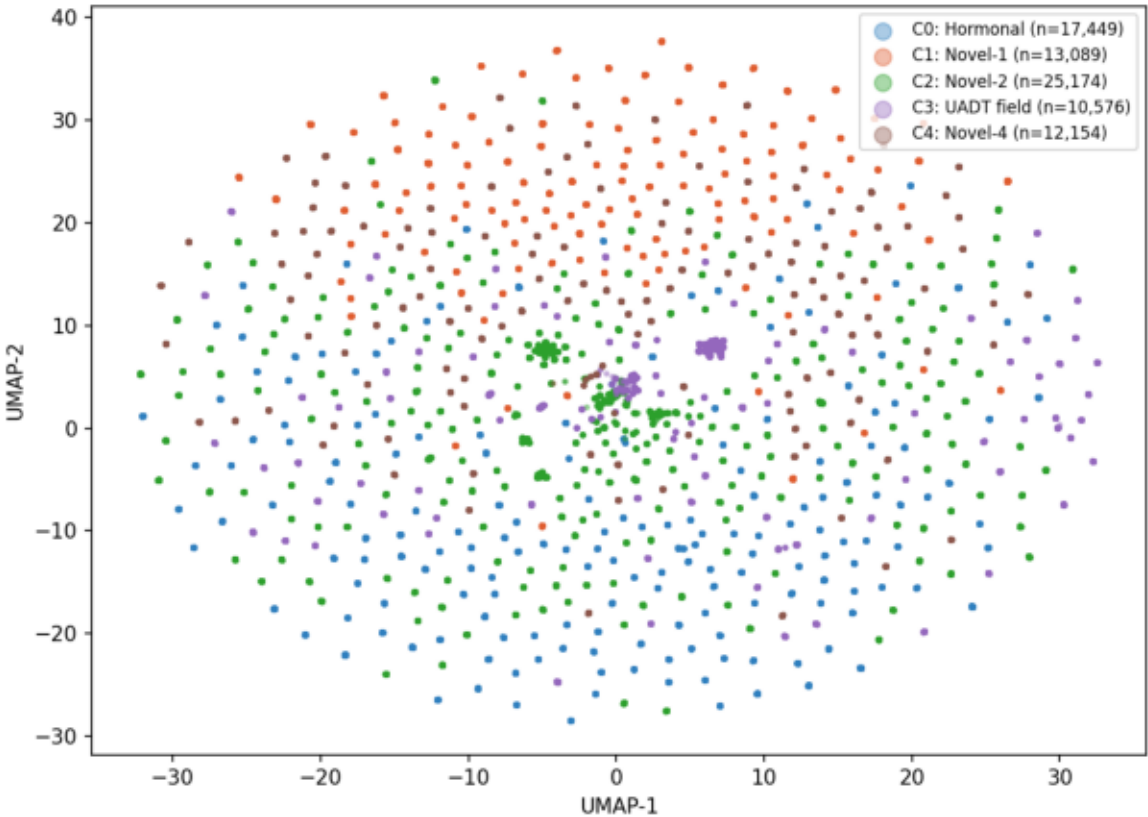
Fig 2c: Per-site reconstruction accuracy



Dim	Axis	Top sites (+)	Max ρ
z0	Hormonal/gynecologic	C18, C20, C61	0.6366026954412773
z1	UADT/field-cancerization	C34, C61, C20	0.422072304559313
z2	Hormonal/gynecologic	C22, C67, C02	0.533871797715667
z3	Liver-GI	C18, C20, C67	0.4852947144479425
z4	UADT/field-cancerization	C50, C53, C20	0.6472972092384158
z5	Hormonal/gynecologic	C61, C18, C20	0.5926193368596674
z6	Liver-GI	C34, C20, C06	0.4476132307563084
z7	UADT/field-cancerization	C34, C22, C61	0.5514953137902937
z8	Hormonal/gynecologic	C53, C20, C50	0.3736223964135032
z9	UADT/field-cancerization	C34, C06, C02	0.3622451959274911
z10	Liver-GI	C22, C61, C34	0.5400280979990673
z11	Hormonal/gynecologic	C18, C20, C53	0.5320601136972001

Fig 3a: UMAP — KMeans clusters

VAE latent space — UMAP projection, KMeans k=5

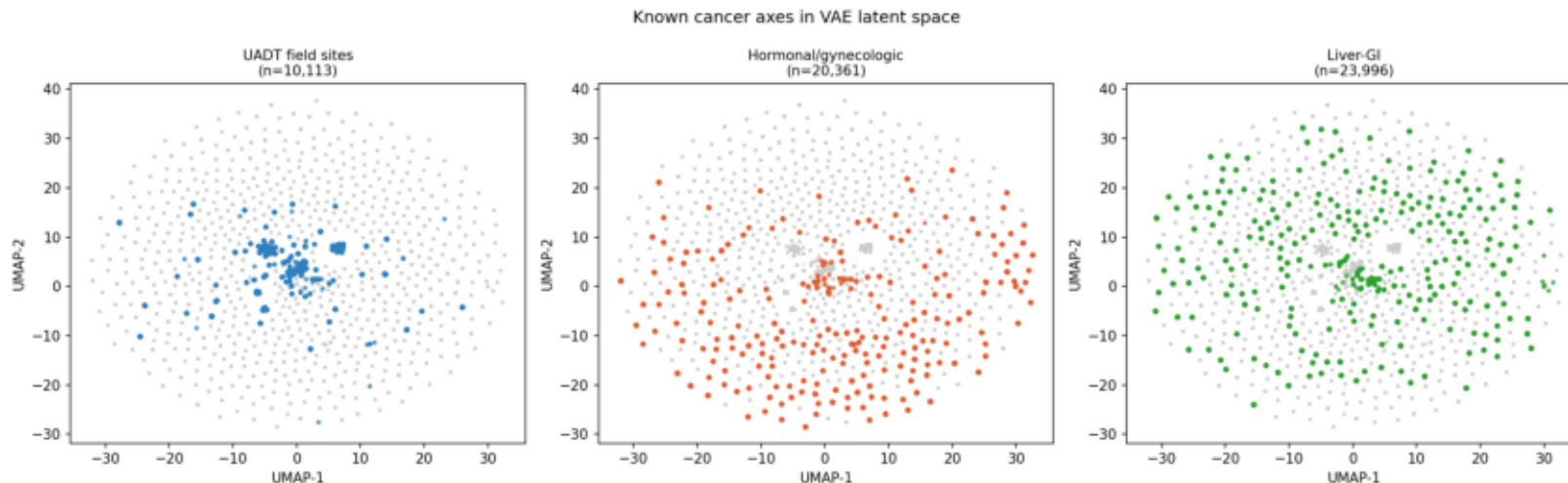


Cluster	Name	N	Top-5 enriched sites
0	Hormonal	17449	C53, C50, C21, C34, C17
1	Novel-1	13089	C34, C61, C71, C22, C17
2	Novel-2	25174	C20, C77, C44, C38, C17
3	UADT field	10576	C54, C56, C06, C16, C17
4	Novel-4	12154	C22, C08, C69, C17, C16

KMeans clustering (k=5) applied to VAE latent μ vectors. Clusters are characterized by fold-enrichment in cancer site prevalence relative to the full registry. Known axes (UADT, hormonal) should map to distinct clusters if the VAE has learned meaningful structure. 'Novel' clusters represent axes not predicted by prior hypotheses and warrant follow-up characterization.

Known Axis Alignment

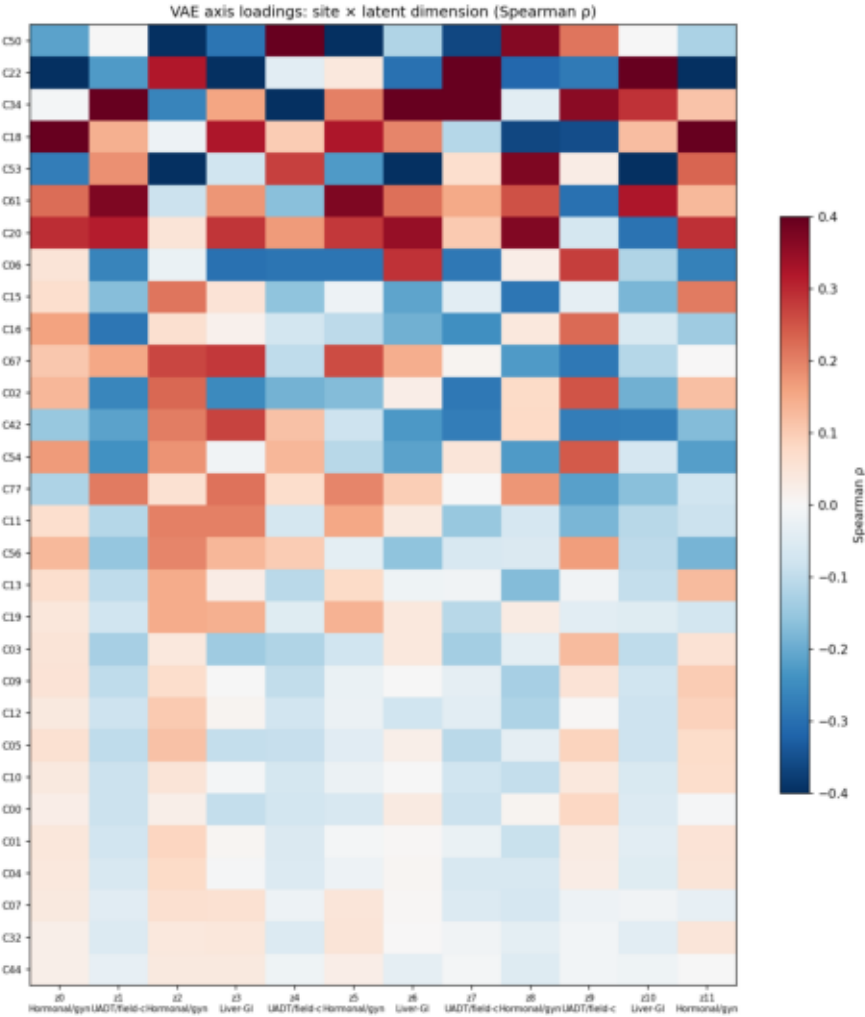
Fig 3b: Known cancer axes projected into VAE latent space
(UADT field sites / Hormonal-gynecologic / Liver-GI)



If VAE has captured biologically meaningful structure, patients with UADT field cancers should cluster separately from hormonal/GI patients. Overlap between known axes and novel clusters points to shared exposures worth investigating (e.g. radiation overlap between thyroid + breast + UADT).

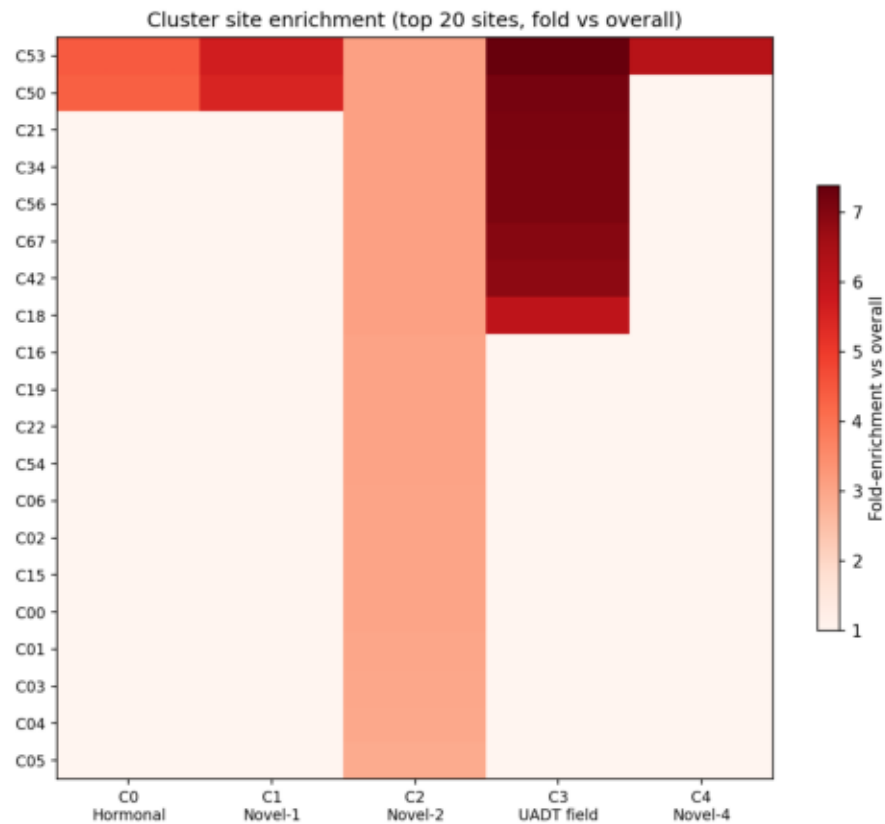
Per-Dimension Axis Loadings

Fig 3c: VAE axis loadings — site × latent dimension (Spearman ρ)
Red = positive loading (site enriched at high z), Blue = negative



Novel Axis Candidates + Limitations

Fig 3d: Cluster × site fold-enrichment



No novel axes detected — all dimensions mapped to known biology.

Limitations:

- ① Registry contains only cancer diagnoses — no molecular, treatment, or lifestyle data; axes reflect co-occurrence patterns not etiology.
- ② Single-cancer patients (the majority) are represented as zero vectors; sparse input may cause VAE to place them all near the prior mean.
- ③ KMeans $k=5$ is arbitrary; silhouette analysis and varied k recommended before publication.
- ④ $\beta=1$ (standard ELBO); increasing β would encourage more axis-disentanglement but risks underfitting rare co-occurrences.
- ⑤ UMAP projection is stochastic — cluster topology varies slightly across runs (seed=42 locked here).
- ⑥ Site MIN_SITE_N=30 threshold removes ultra-rare cancers; results do not generalise to those.